

3D GIA models

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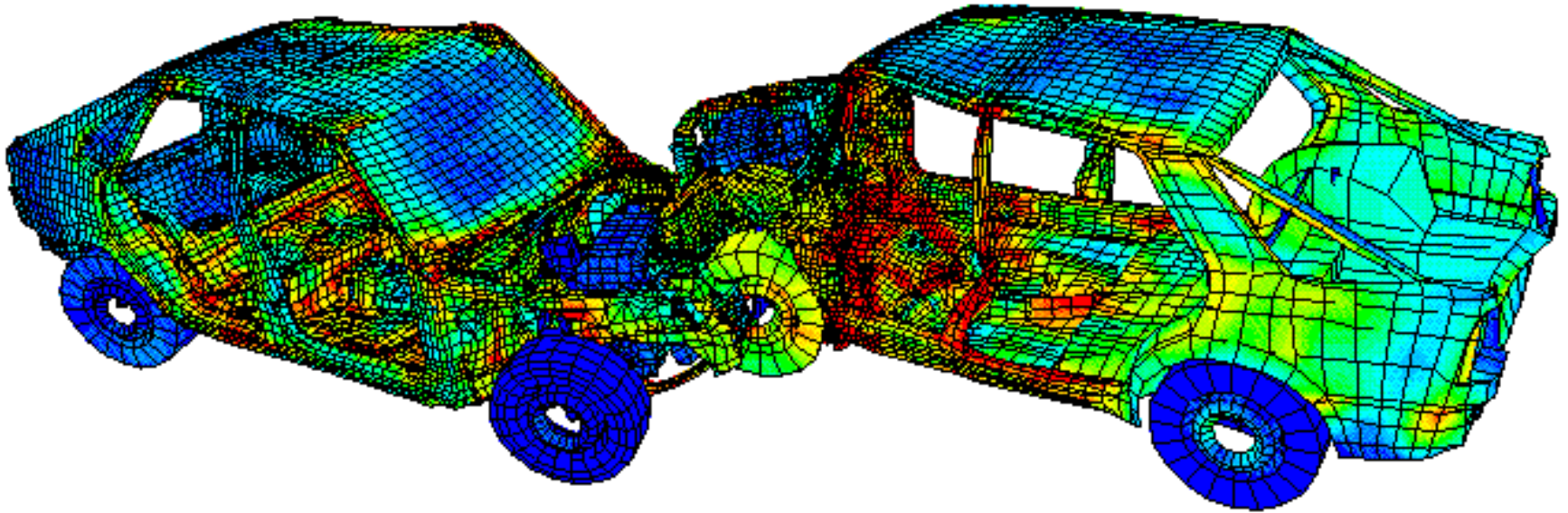
With contributions from:

Bas Blank, Rosalie van Casteren, Tim Hermans, Haiyang Hu, Dirk Oude Egbrink, Maaïke Weerdesteijn

Contents

- 3D Modelling: Finite-Element method (18 slides)
- Overview of different methods (10 slides)
- Applications of 3D GIA models (16 slides)

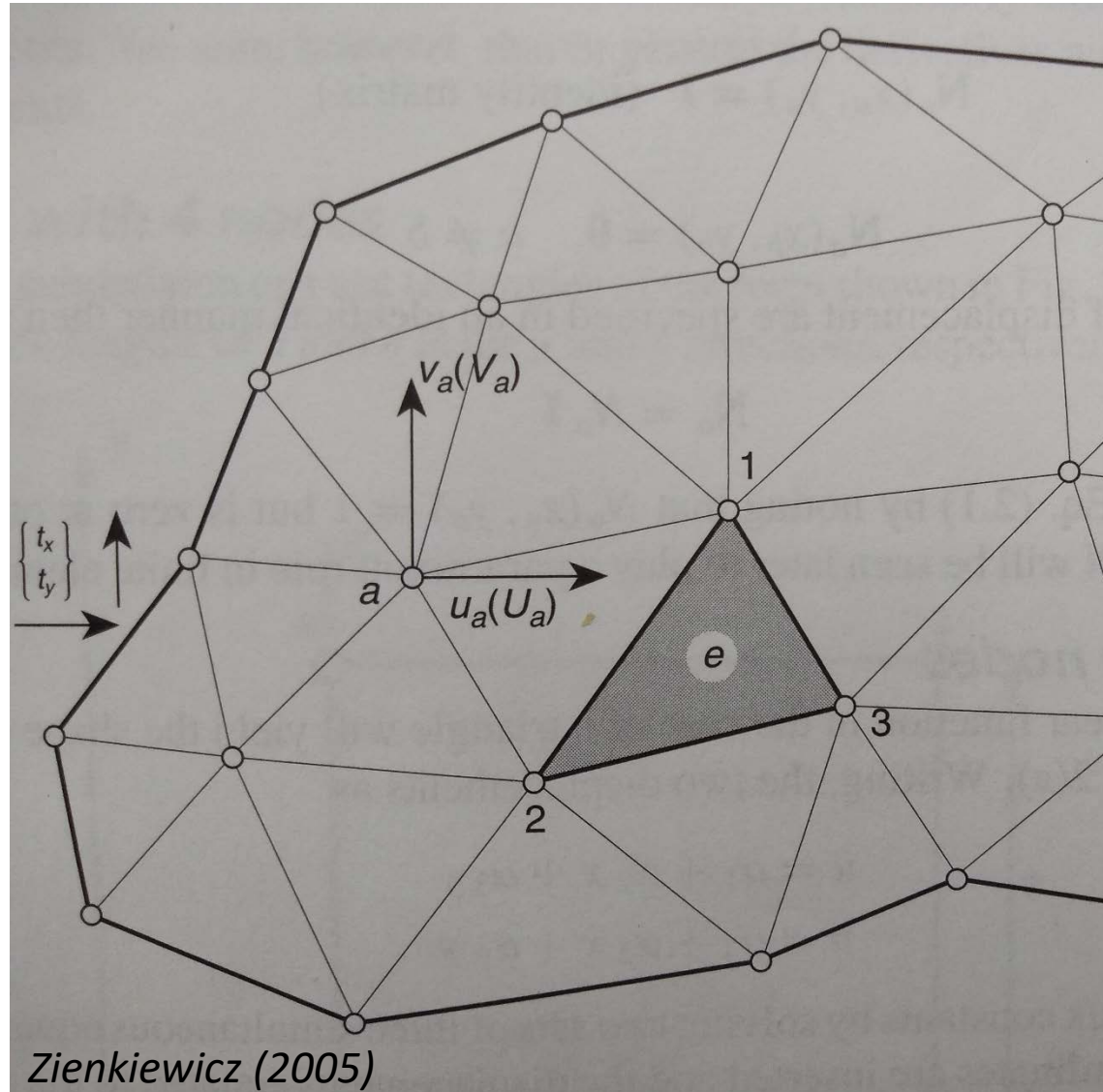
Finite Element Method



Finite Element Method

based on *Zienkiewicz (2005)*

1. Elements e
2. Node displacement u_a, v_a
3. $u(x, y) = f(u_a, v_a)$
4. $u, v \rightarrow$ strain ϵ
5. $\epsilon \rightarrow$ stress σ
6. $\mathbf{q}_a$ force equilibrium



Finite Element Method

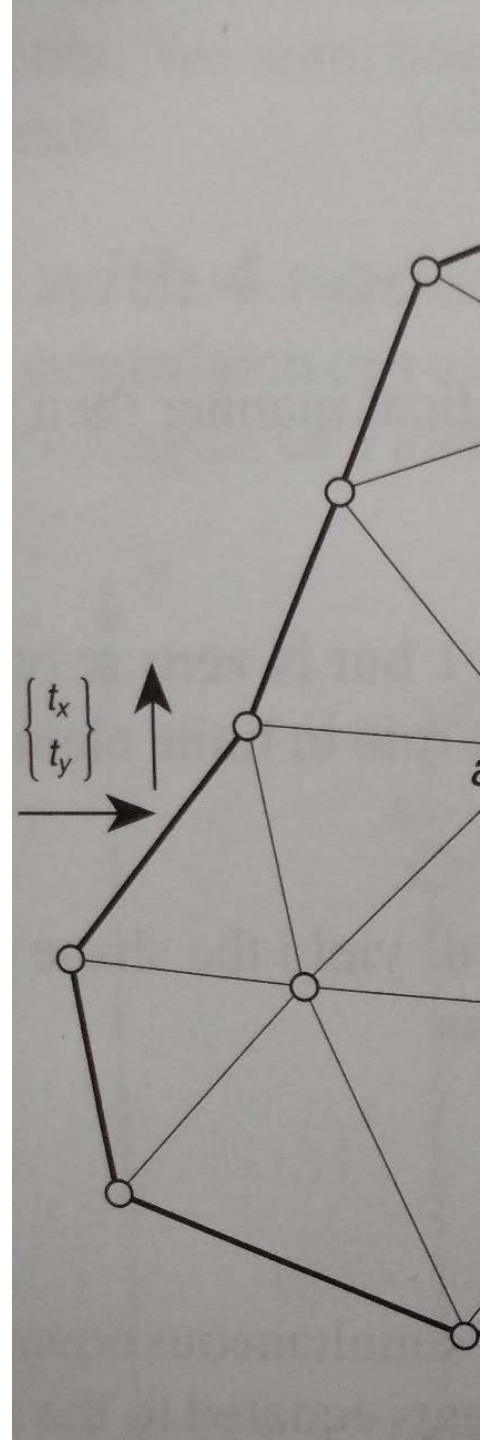
$$\int_V \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} dV = \int_V \delta \mathbf{u}^T \mathbf{b} dV + \int_S \delta \mathbf{u}^T \mathbf{t} dS$$

t is traction or stress vector

Principal of Virtual Work :

work done by the internal stresses =

work done by the body forces and tractions



Finite Element Method

Conservation of momentum (**b** is a body force)

$$\nabla \cdot \boldsymbol{\sigma} - \mathbf{b} = 0$$

multiplying with a test function

$$\delta \mathbf{u}^T = (\delta u \quad \delta v \quad \delta w)$$

and integrating over the whole domain

$$\int_V \delta \mathbf{u}^T (\nabla \cdot \boldsymbol{\sigma} - \mathbf{b}) dV = 0$$

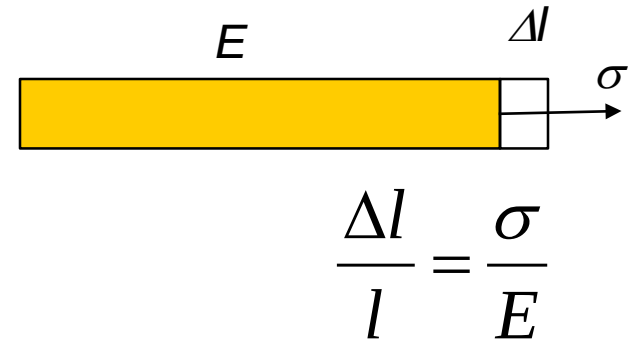
integrating by parts and rearranging

$$\int_V \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} dV = \int_V \delta \mathbf{u}^T \mathbf{b} dV - \int_S \delta \mathbf{u}^T \mathbf{t} dS$$

Stress-strain relation

$$\boldsymbol{\sigma} = \mathbf{D} \delta \boldsymbol{\varepsilon}$$

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{bmatrix} \quad \delta \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{bmatrix}$$



$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & \nu & 0 & 0 & 0 \\ \nu & (1-\nu) & \nu & 0 & 0 & 0 \\ \nu & \nu & (1-\nu) & 0 & 0 & 0 \\ 0 & 0 & 0 & (1-2\nu)/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & (1-2\nu)/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & (1-2\nu)/2 \end{bmatrix}$$

Strain to displacement

$\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon}$ Substituting in the equation of virtual work:

$$\int_V \delta \boldsymbol{\varepsilon}^T \mathbf{D} \boldsymbol{\varepsilon} dV = \int_V \delta \mathbf{u}^T \mathbf{b} dV + \int_S \delta \mathbf{u}^T \mathbf{t} dS$$

$$\delta \boldsymbol{\varepsilon} = \mathbf{S} \delta \mathbf{u}$$

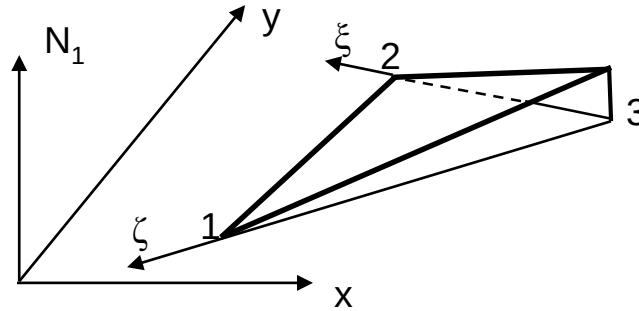
$$\delta \boldsymbol{\varepsilon} = \begin{bmatrix} \delta \varepsilon_x \\ \delta \varepsilon_y \\ \delta \varepsilon_z \\ \delta \gamma_{xy} \\ \delta \gamma_{yz} \\ \delta \gamma_{zx} \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 \\ 0 & \frac{\partial}{\partial y} & 0 \\ 0 & 0 & \frac{\partial}{\partial z} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \end{bmatrix}$$

$$\delta \mathbf{u} = \begin{pmatrix} \delta u \\ \delta v \\ \delta w \end{pmatrix}$$

Shape functions

$$\delta \mathbf{u} \approx \mathbf{N} \delta \tilde{\mathbf{u}} = \mathbf{N} \begin{pmatrix} \delta \tilde{u}_a \\ \delta \tilde{v}_a \\ \delta \tilde{w}_a \end{pmatrix}$$



$\delta \tilde{\mathbf{u}}$ is the virtual displacement at the nodes of the element

$\mathbf{N}$ are the *shape functions*. Find nodal parameters such that N equals 1 at the node you are interested in

$$\mathbf{N} = \frac{1}{2\Delta} \begin{bmatrix} x_2 y_3 - x_3 y_2 + (y_2 - y_3)x + (x_3 - x_2)y & \dots & \dots \end{bmatrix}$$

$$\delta \boldsymbol{\varepsilon} = \mathbf{S} \begin{pmatrix} \delta u \\ \delta v \\ \delta w \end{pmatrix} \approx \mathbf{S} \mathbf{N} \delta \tilde{\mathbf{u}} = \mathbf{B} \delta \tilde{\mathbf{u}}$$

$$\int_V \delta \boldsymbol{\varepsilon}^T \mathbf{D} \boldsymbol{\varepsilon} dV = \int_V \delta \mathbf{u}^T \mathbf{b} dV + \int_S \delta \mathbf{u}^T \mathbf{t} dS$$

$$\int_V \delta \tilde{\mathbf{u}}^T \mathbf{B}^T \mathbf{D} \mathbf{B} \tilde{\mathbf{u}} dV = \int_V \delta \tilde{\mathbf{u}}^T \mathbf{N}^T \mathbf{b} dV + \int_S \delta \tilde{\mathbf{u}}^T \mathbf{N}^T \mathbf{t} dS$$

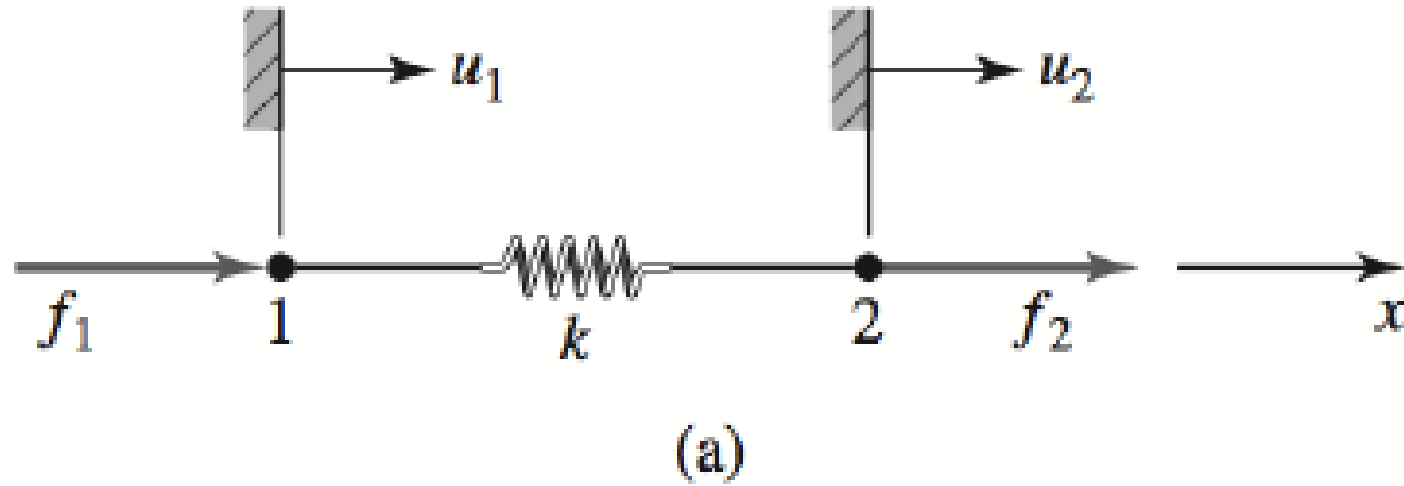
$$\int_V (.) dV = \sum_e \int_{V_e} (.) dV_e \quad \text{Remembering that } \delta \tilde{\mathbf{u}}^T \text{ is arbitrary}$$

$$\mathbf{K}_{ab} = \sum_e \int_{V_e} \mathbf{B}_a^T \mathbf{D} \mathbf{B}_b dV \quad \text{Stiffness matrix}$$

$\mathbf{K}_{ab} \tilde{\mathbf{u}}_b + \mathbf{f}_a = 0$

$$\mathbf{f}_a = - \sum_e \int_{V_k} \mathbf{N}_a \mathbf{b} dV - \sum_e \int_{S_e^t} \mathbf{N}_a \mathbf{t} dS_e^t \quad \text{Load matrix}$$

Exercise



Example taken from a pdf of MECH410 at the University of Victoria

Assuming that both the nodal displacements are zero when the spring is undeformed, the net spring deformation is given by

$$\delta = u_2 - u_1$$

and the resultant axial force in the spring is

$$f = k\delta = k(u_2 - u_1)$$

For equilibrium,

$$f_1 + f_2 = 0 \text{ or } f_1 = -f_2,$$

Then, in terms of the applied nodal forces as

$$f_1 = -k(u_2 - u_1)$$

$$f_2 = k(u_2 - u_1)$$

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which can be expressed in matrix form as

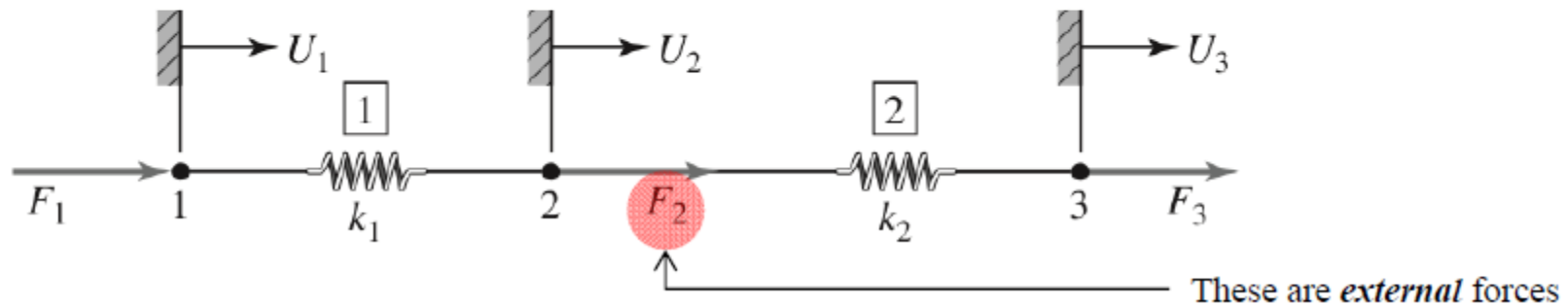
$$\begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} f_1 \\ f_2 \end{Bmatrix} \quad \text{or} \quad [k_e]\{u\} = \{f\}$$

where

$$[k_e] = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$$

Stiffness matrix for one spring element

SYSTEM OF TWO SPRINGS



$$\begin{bmatrix} k_1 & -k_1 & 0 \\ -k_1 & k_1 + k_2 & -k_2 \\ 0 & -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \\ U_3 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \end{Bmatrix}$$

Continue building the stiffness matrix, solve by least-squares

GIA into the FE framework

1. Conservation of momentum (linearized, inertial force neglected)

$$\begin{array}{c} \text{Self-gravitation} \end{array} \quad \begin{array}{c} \text{Advection of} \\ \text{hydrostatic} \\ \text{pressure} \end{array}$$

$$\nabla \cdot \boldsymbol{\sigma} - \rho_0 \nabla \phi_1 - \rho_1 g_0 \mathbf{e}_r - \nabla (\mathbf{u} \cdot \rho_0 g_0 \mathbf{e}_r) = 0 \quad \Rightarrow \quad \nabla \cdot \boldsymbol{\sigma} - \mathbf{b} = 0$$

Change in stress

0: background state
1: perturbation

Wu (GJI 2004):

stress transformation

$$\mathbf{t} = \boldsymbol{\sigma} - (\rho_0 g_0 u_r + \rho_0 \phi_1) \mathbf{I}$$

Gasperini and Sabadini (GJI 1989):

write buoyancy force as a volume integral times displacement and add to the stiffness matrix

[https://www.youtube.com/watch?v= Kv7p1xjnd8](https://www.youtube.com/watch?v=Kv7p1xjnd8)

Where does viscosity enter?

Zhong et al. (2003)

$$\dot{\varepsilon}_{ij}^v = \frac{1}{2\eta}(\sigma_{ij} + P\delta_{ij}),$$

$$\varepsilon_{ij}^e = \frac{1}{2\mu}(\sigma_{ij} + P\delta_{ij}),$$

$$\sigma_{ij} + \frac{\eta}{\mu}\dot{\sigma}_{ij} = -\left(P + \frac{\eta}{\mu}\dot{P}\right)\delta_{ij} + 2\eta\dot{\varepsilon}_{ij}.$$



Maxwell

Use some kind of numerical integration to compute stress at the next times step

$$\sigma_{ij}^n = -P^n\delta_{ij} + \frac{2\eta}{\alpha + \Delta t/2}\Delta\varepsilon_{ij}^n + \frac{\alpha - \Delta t/2}{\alpha + \Delta t/2}(\sigma_{ij}^{n-1} + P^{n-1}\delta_{ij})$$

Coupling to ice model!

2. Poisson equation

$$\nabla^2 \phi = 4\pi G \rho_1$$

Compute displacement from FE

Compute gravitational potential

Compute new boundary conditions for the FE model

Compute new displacements

...

The model is said to be *self-gravitating*

Iteration can be done at the same time as computing sea level

Compressibility

$$\nabla \cdot \boldsymbol{\sigma} - \rho_0 \nabla \phi_1 - \rho_1 g_0 \mathbf{e}_r - \nabla (\mathbf{u} \cdot \rho_0 g_0 \mathbf{e}_r) = 0$$

$$\nabla^2 \phi = 4\pi G \rho_1$$

$$\rho_1 = -\rho_0 \nabla \cdot \mathbf{u} - \mathbf{u} \frac{\partial \rho}{\partial r} \mathbf{e}_r$$

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & \nu \\ \nu & (1-\nu) & \nu \\ \nu & \nu & (1-\nu) \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

How to deal with compressibility (changing density of the material?)

Compute the volume change of each element

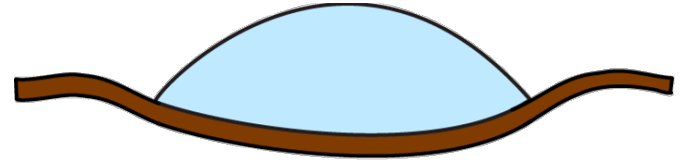
A et al. (GJI 2013)

Wong et al (GJI 2018)

Boundary conditions

Stress (Surface of the Earth)

$$\sigma_{radial} = -\rho_{ice} h_{ice} g_0; \quad \sigma_{horizontal} = 0$$



Stress (Internal boundaries)

$$\sigma_{radial} \Big|_{-}^{+} = -\rho (g_0 u_{radial} - \phi_1)$$

Winkler foundation

Displacement (Internal boundaries)

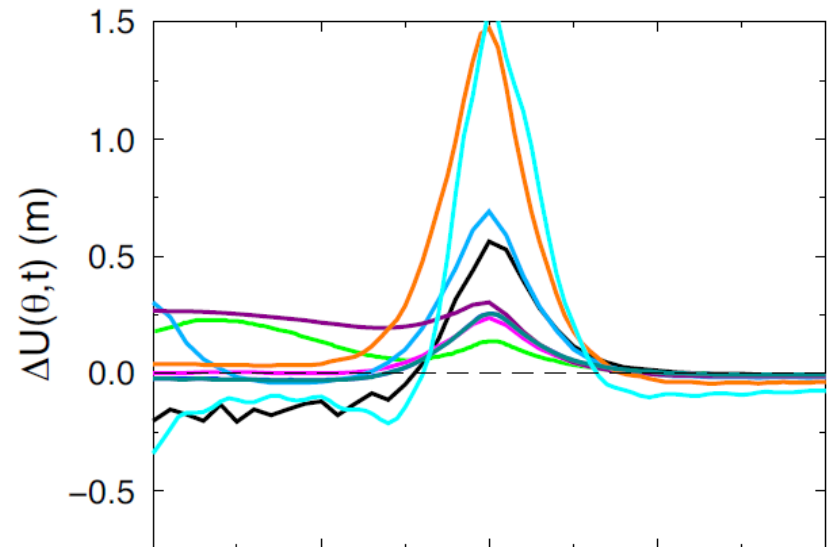
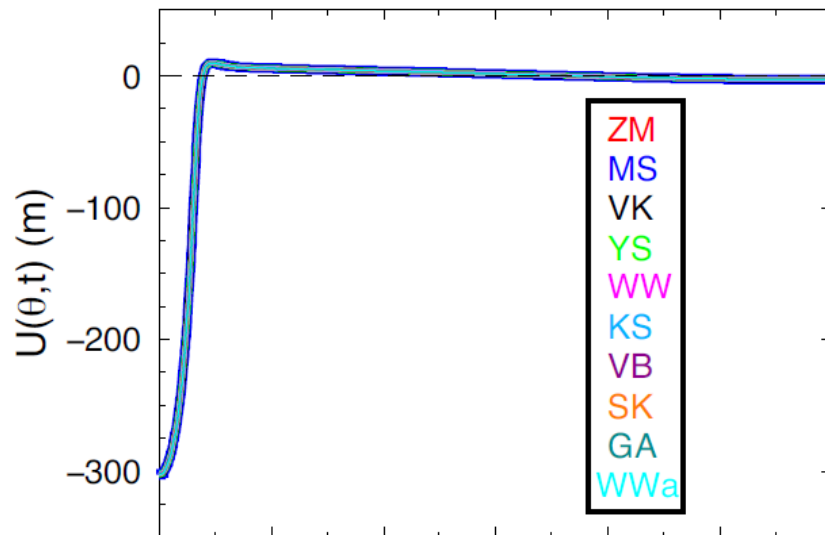
$$[\mathbf{u}]_{-}^{+} = 0$$

Gravitational gradient and its derivative

...

Benchmark

Martinec et al (2018)



No comparison yet between 3D models with 3D rheology!

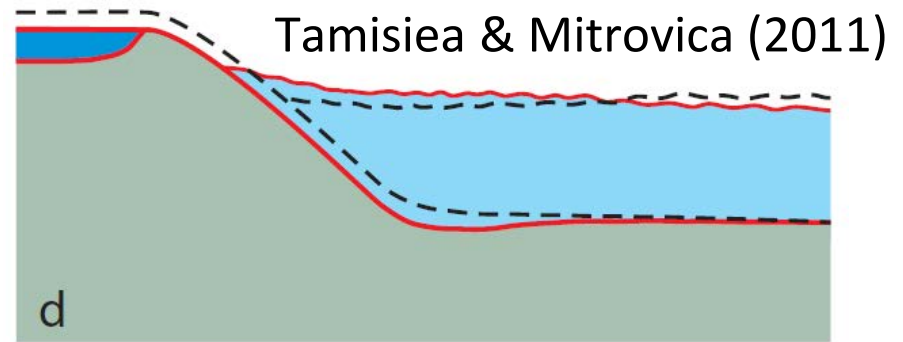
3D models - Practical differences

- Spherical or flat?
- Self-gravitating
- Compressible
- Commercial
- Accuracy: Size of elements, shape functions, time integration
- Boundary conditions

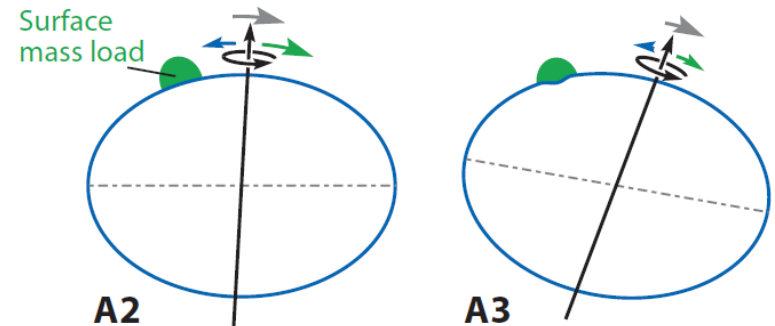
Other effects

Sea level

Different resolution!

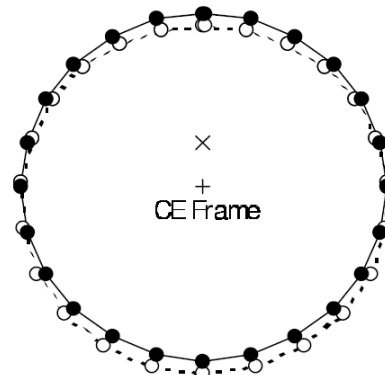


Rotational feedback



Mitrovica and Wahr (2011)

Geocenter motion



Blewitt (2003)

Spectral-finite element method

Martinec (2000)

Spherical harmonics in the horizontal direction, finite-elements in the radial direction

Sea-level equation: Hagedoorn et al. (2007)

Tanaka et al. (2011): compressible

Used by: Klemann et al (JGR 2007)

Computation time ~days

CITCOM

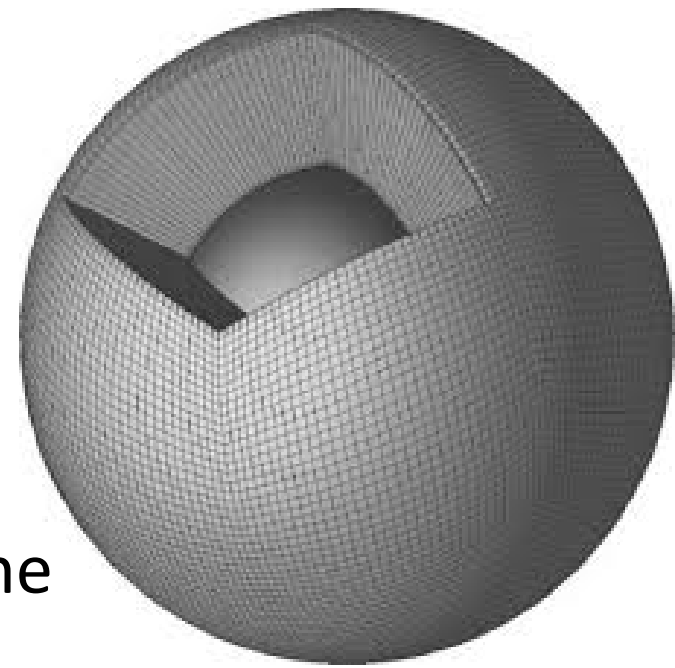
Zhong et al. (2003)

Viscous $\rightarrow$ visco-elastic

Rheological equation depends on time

Paulson et al (2005); sea level equation, rotation

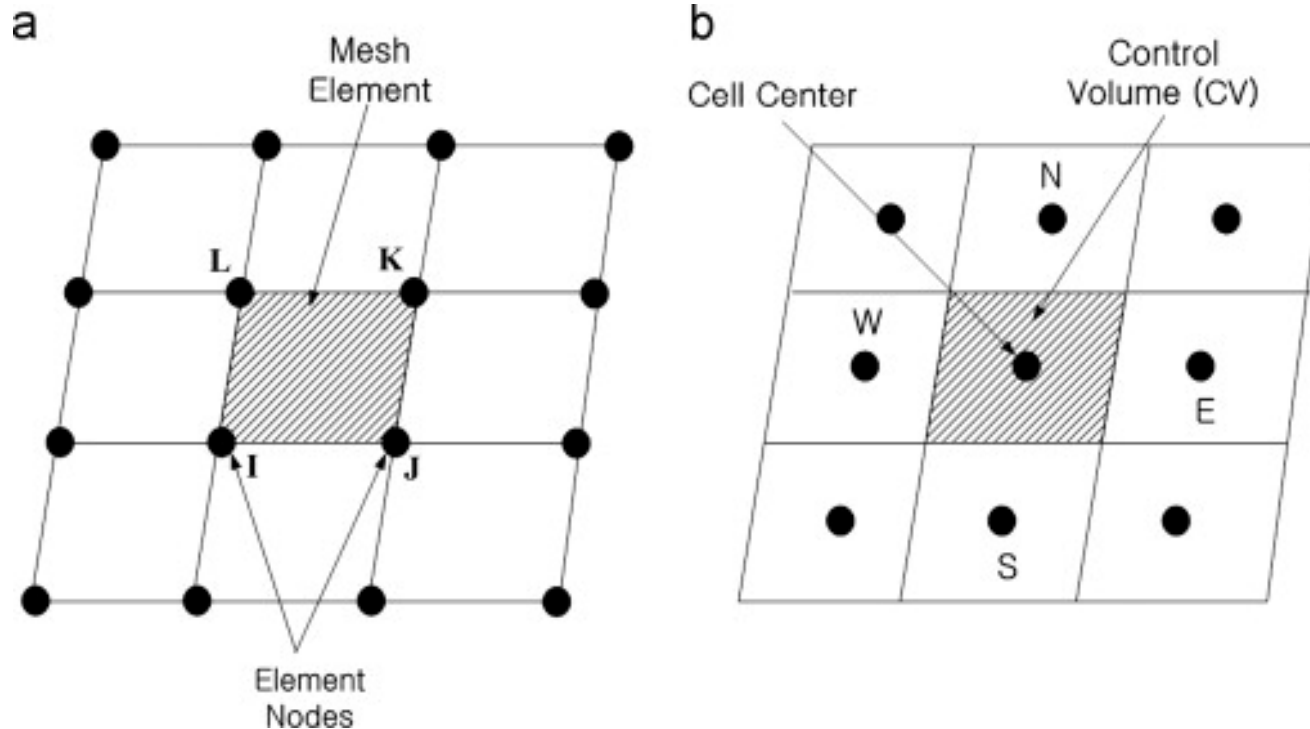
A et al. (2013): compressible



University of Colorado
Boulder

Finite-volume approach

Finite-volume approach (Latychev et al. 2005)



FV handles non-conforming meshes more easily

Used by Austermann et al. (2013), Gomez et al. (2018), Milne et al. (2018)

Wu (2004)



UNIVERSITY OF
CALGARY

Wu and van der Wal (2003): sea-level equation

Wang and Wu (2006), van der Wal et al (2013), Whitehouse

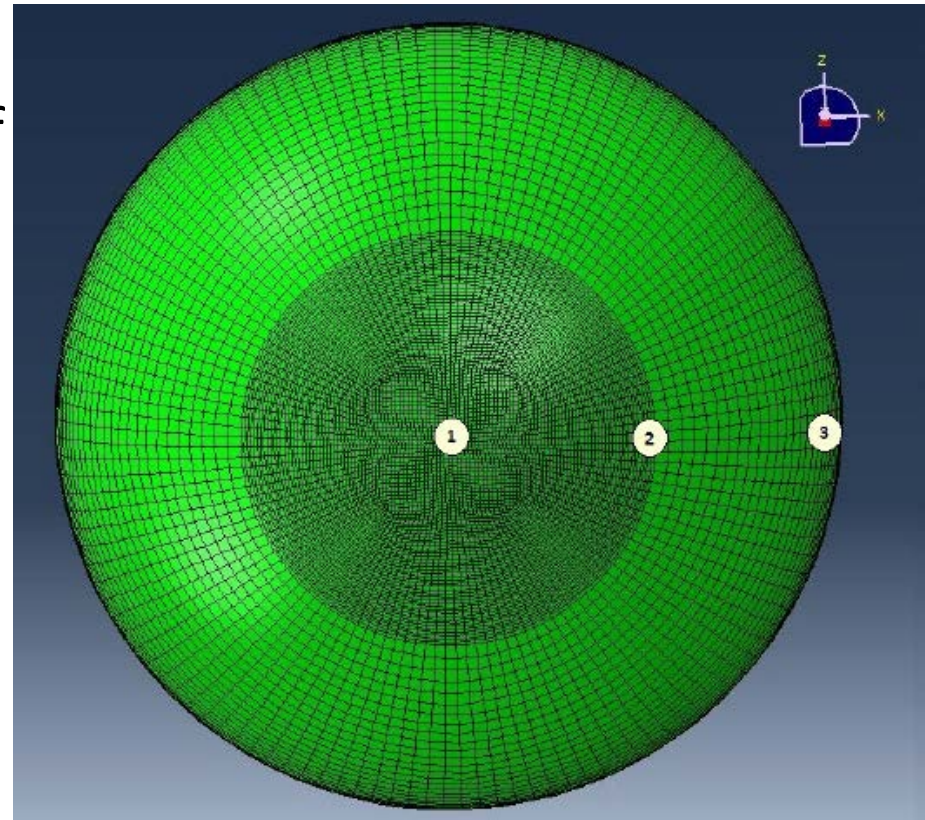
Computation time ~ 2 days

Relatively easy to develop yourself

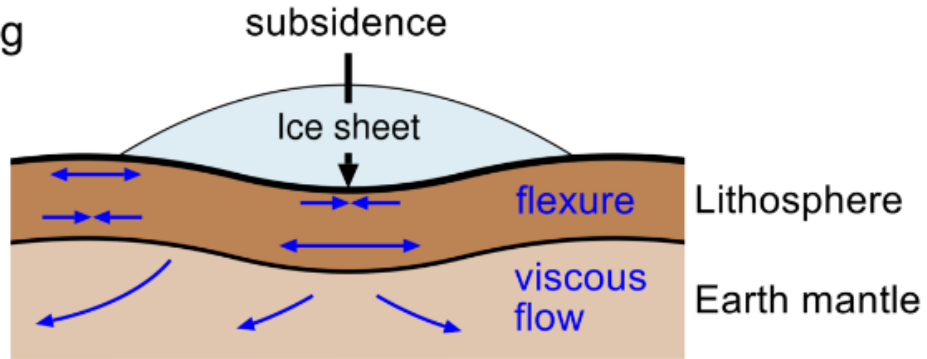
Schotman et al. (PEPI 2008),

Steffen et al (2014)

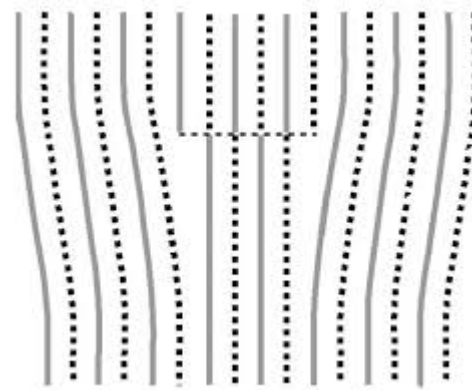
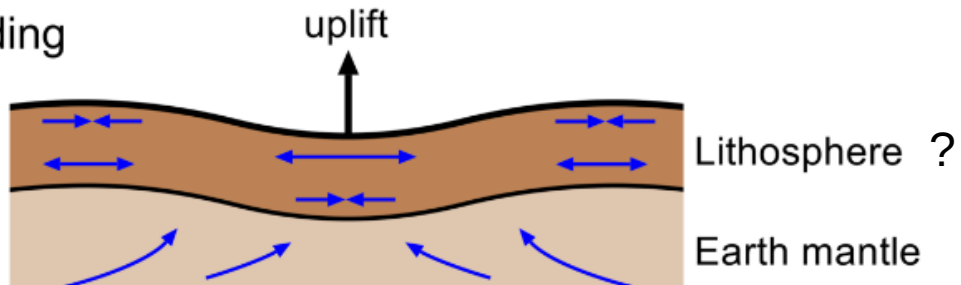
Hu et al (2017) / Bas Blank



Loading



Deloading



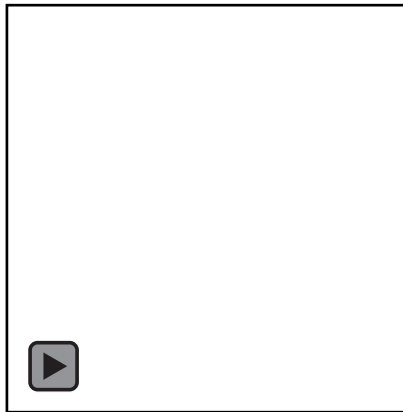
Cordier (2002)

Dislocation creep $\dot{\varepsilon} = f(\sigma^{3.5}, \text{grain size}, \text{wet/dry}, \text{temp})$

Diffusion creep $\dot{\varepsilon} = f(\sigma, \text{grain size}^3, \text{wet/dry}, \text{temp})$

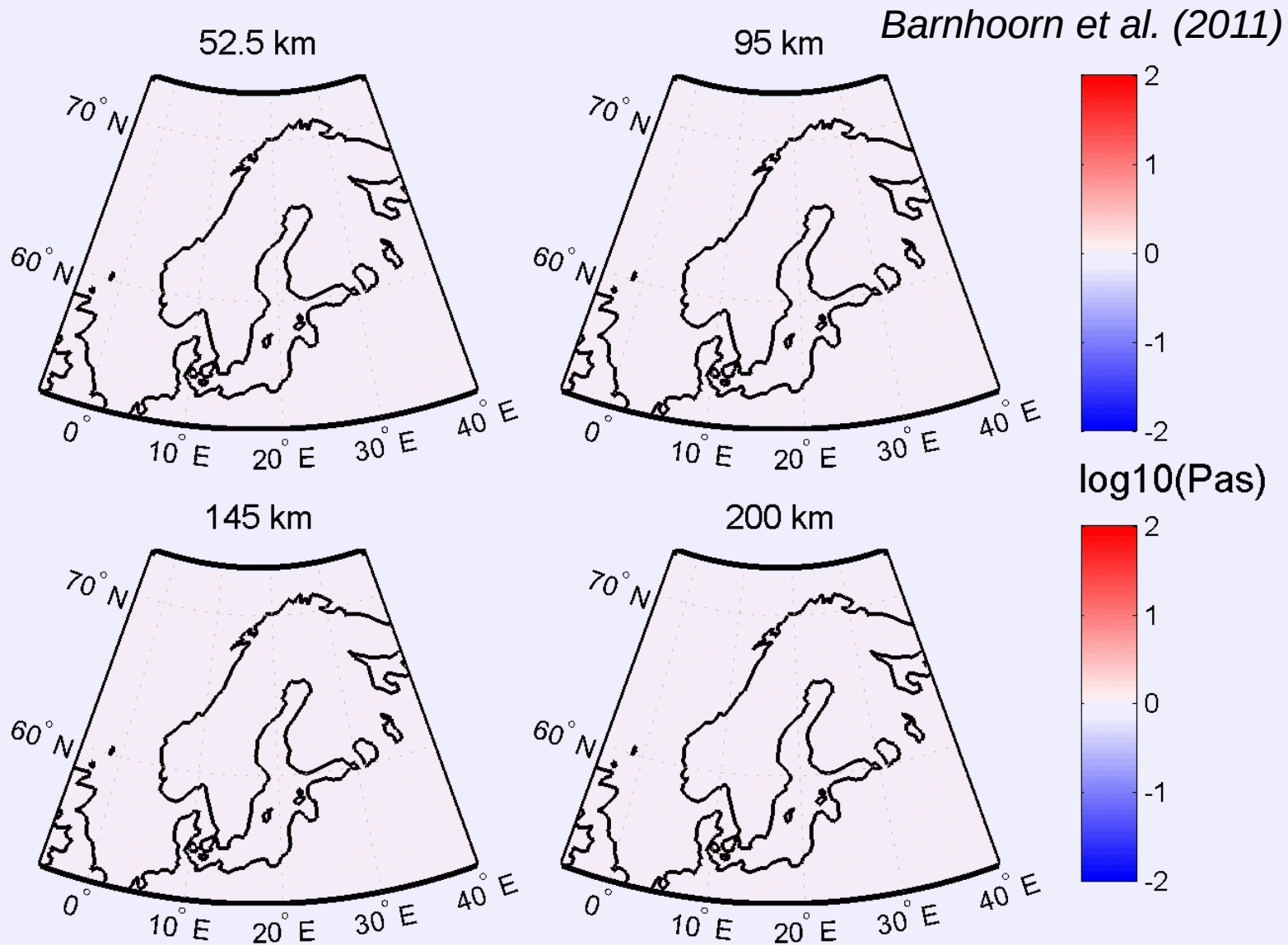
4D model!!!!

$$\eta_{eff} = \frac{\sigma}{2\dot{\varepsilon}}$$



4D viscosity

Effective viscosity change wrt 30,000 yrs before present (wet/10 mm)

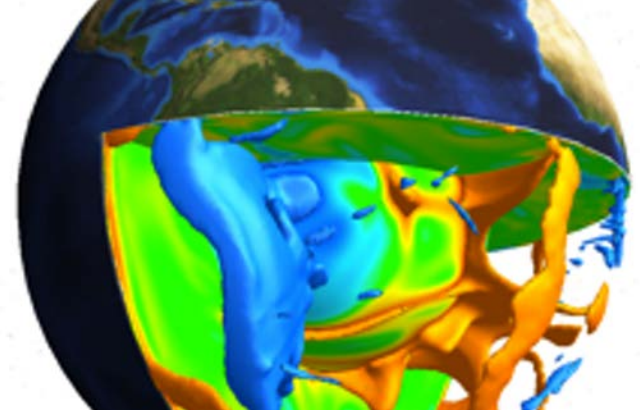


Applications of 3D models

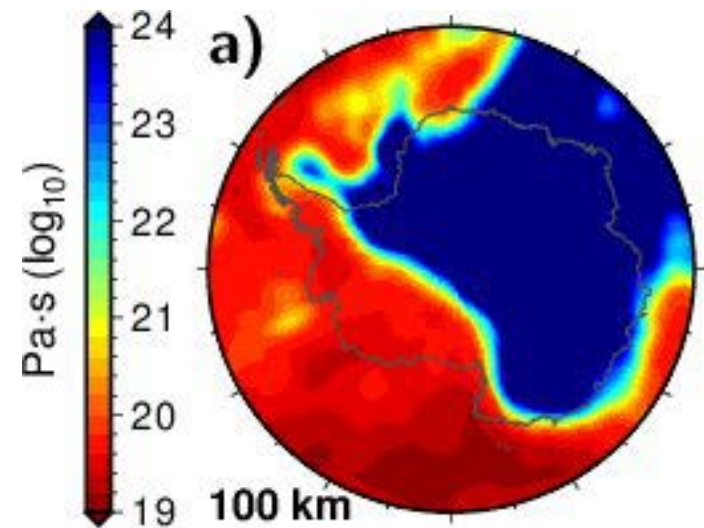
3D takes a long time, where do we need it?

- Constraints on earth rheology, otherwise only spatial and temporal average of viscosity

- Corrections that can not be reproduced by (a combination of) 1D models

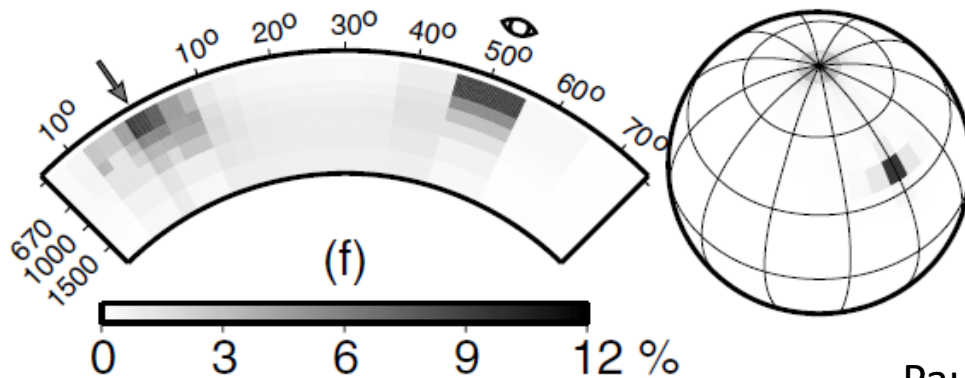
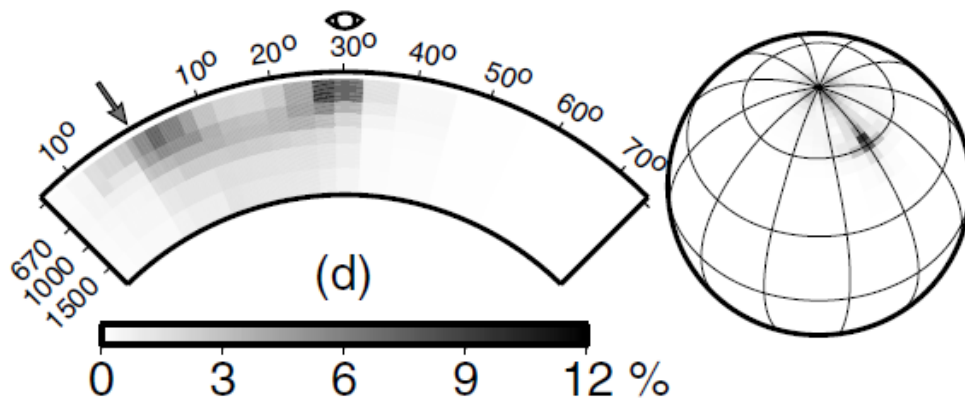
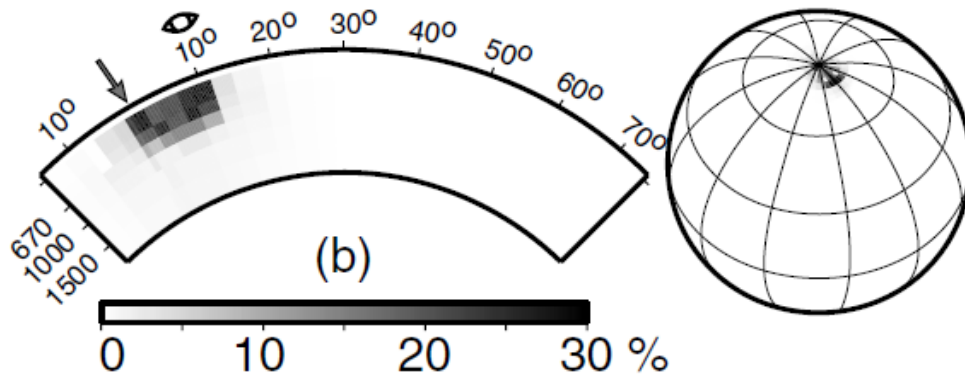


Cardiff University

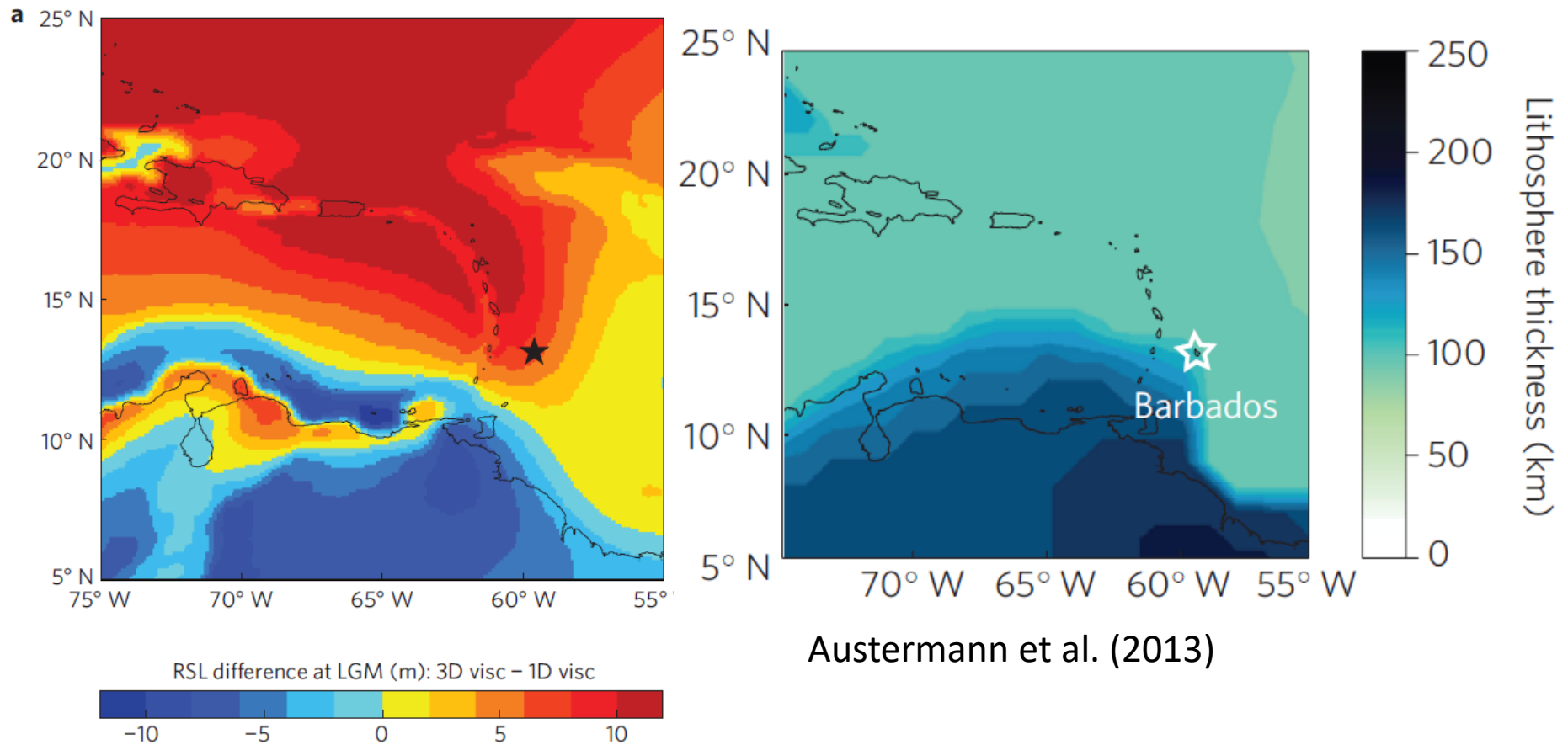


Pappa et al (in review)

3D viscosity - sensitivity

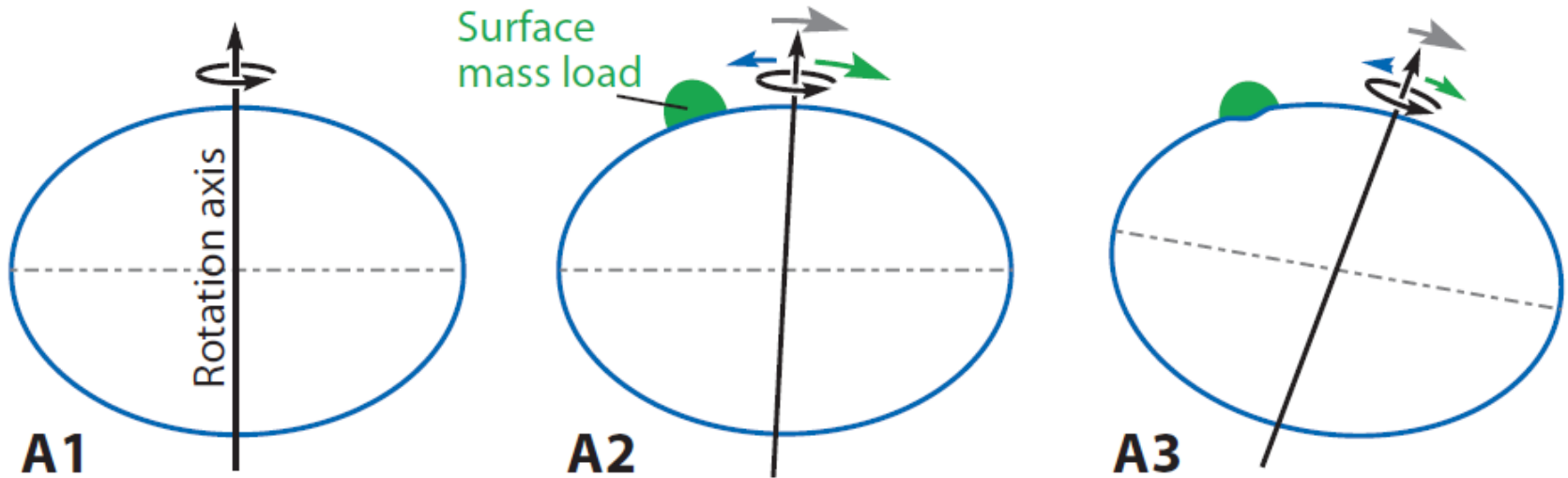


Far-field



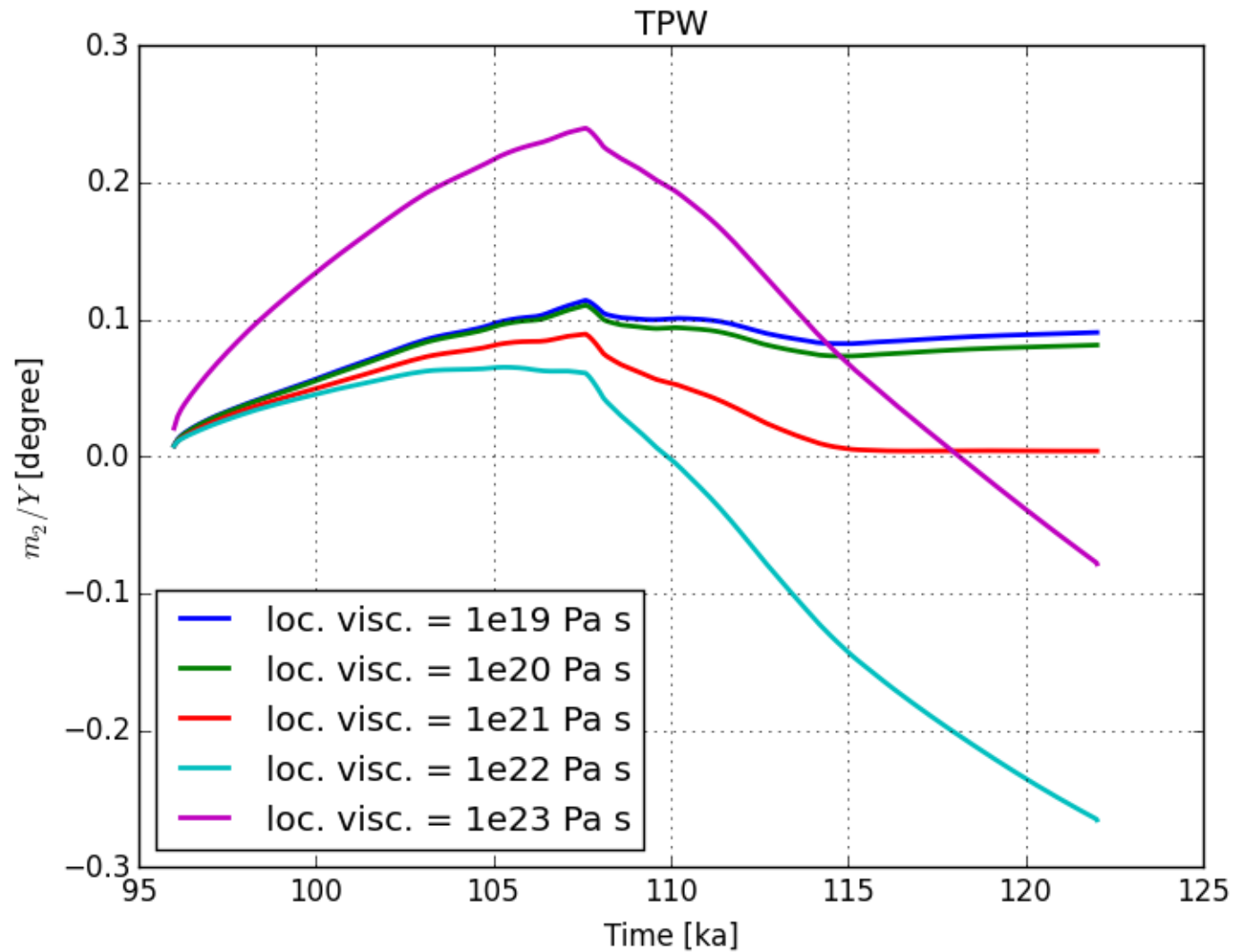
Austermann et al. (2013)

Polar wander



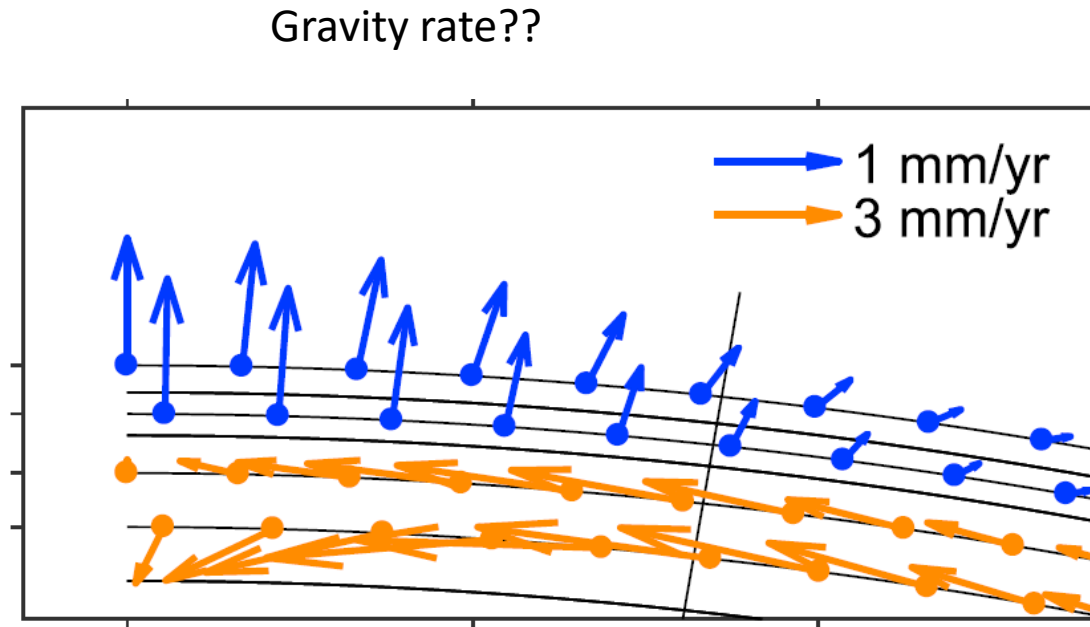
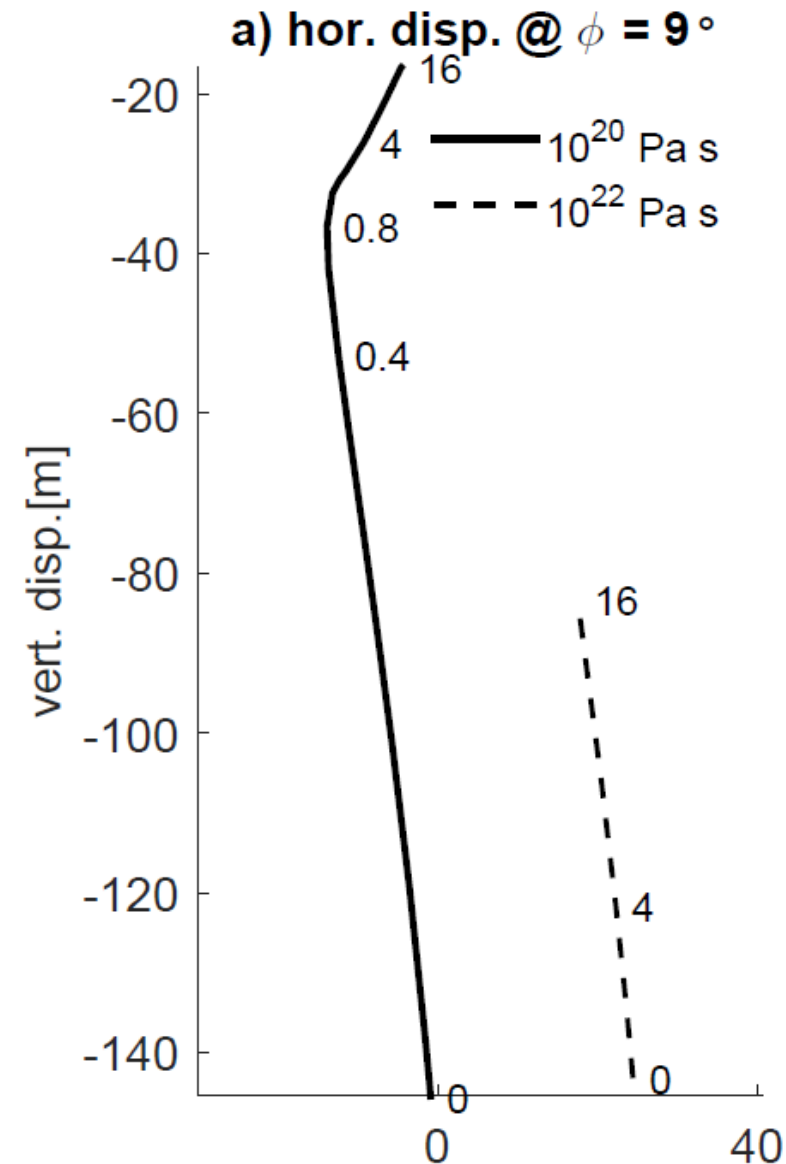
Mitrovica and Wahr (2011)

Polar wander



Maaïke Weerdesteijn

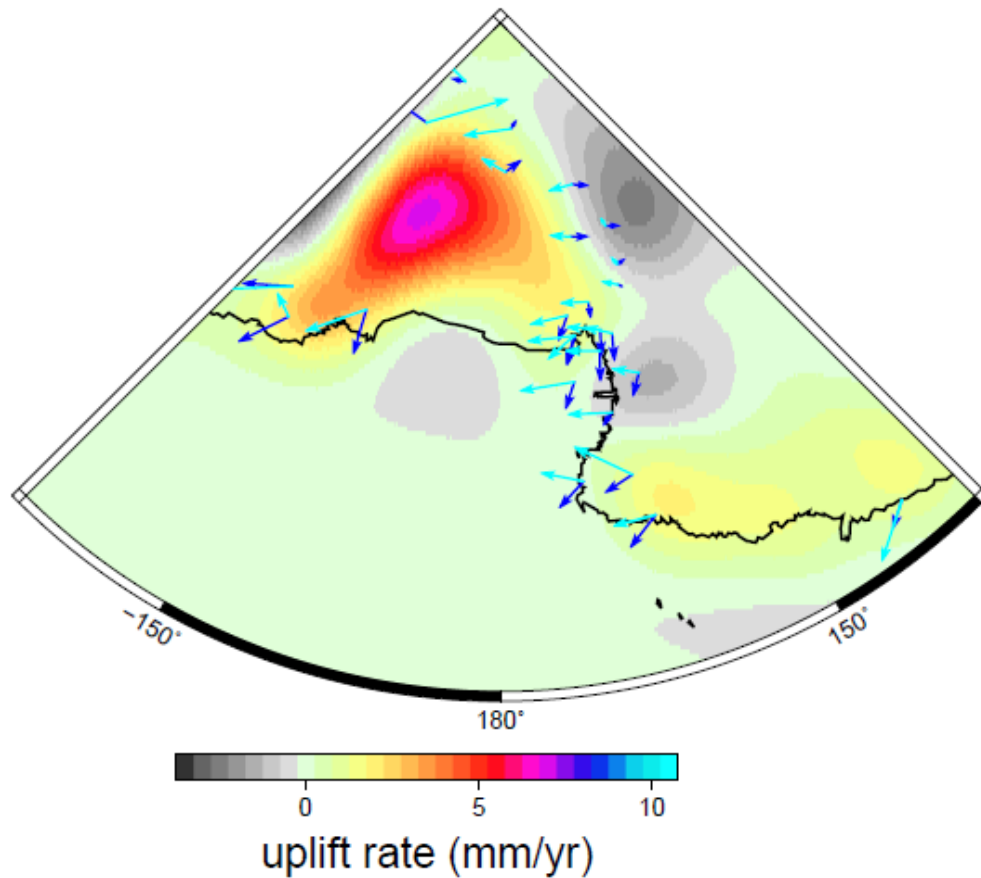
Horizontal velocities



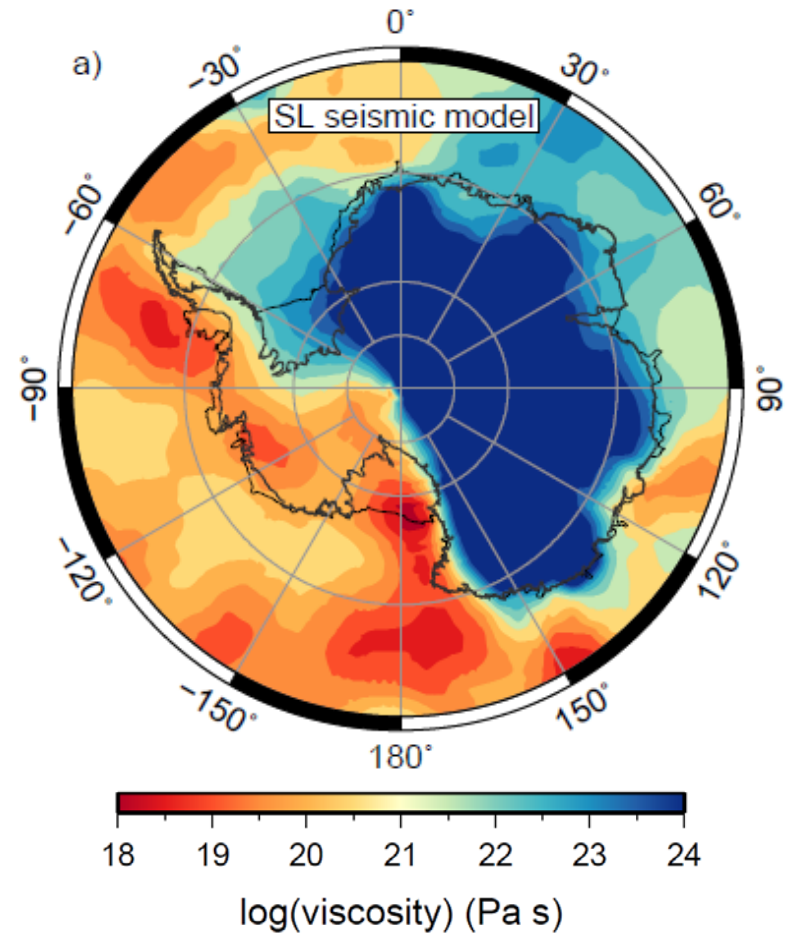
Hermans et al. (GRL 2018)

See also: James and Morgan (GRL 1990)

Horizontal velocities

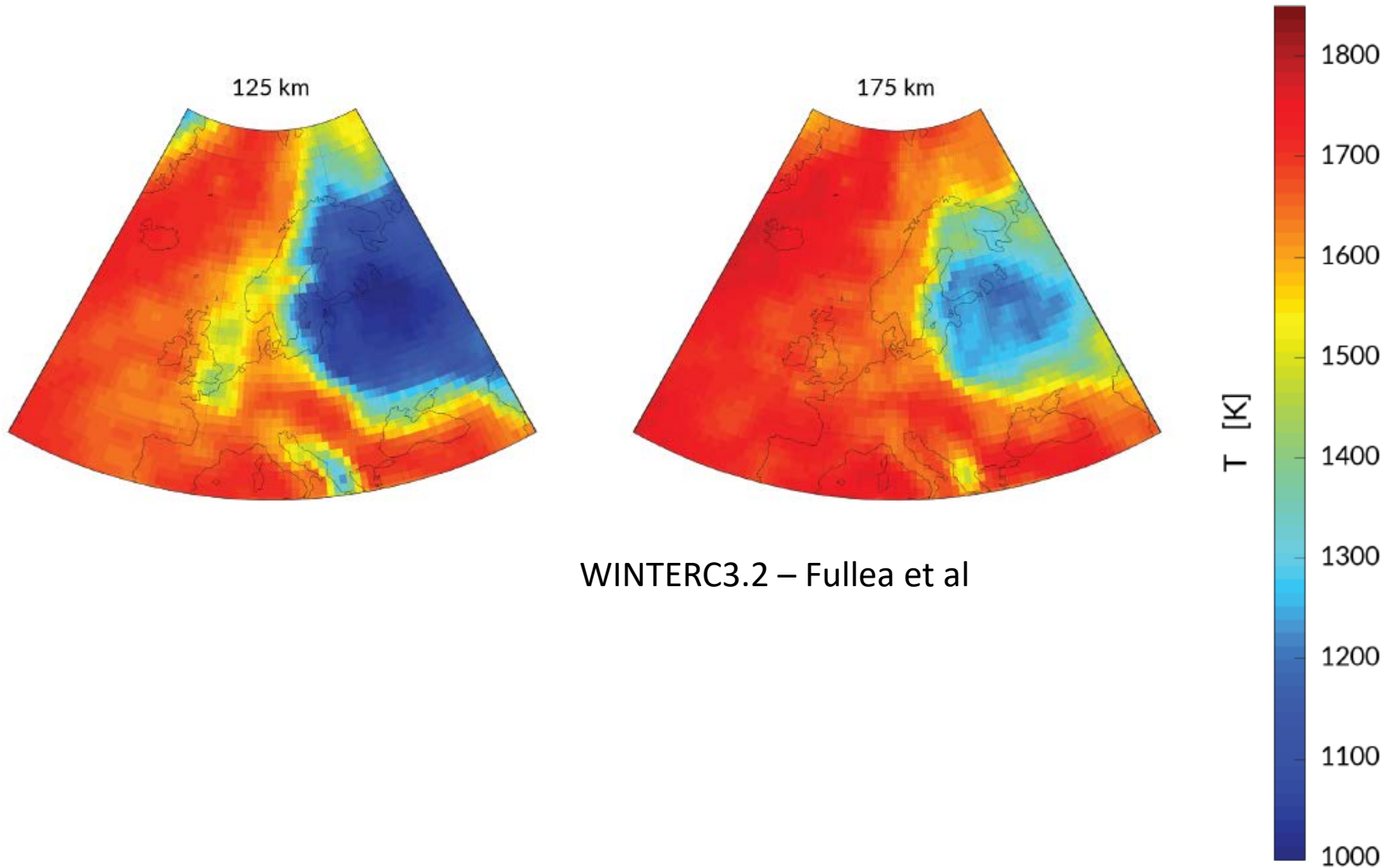


Stephanie Konfal

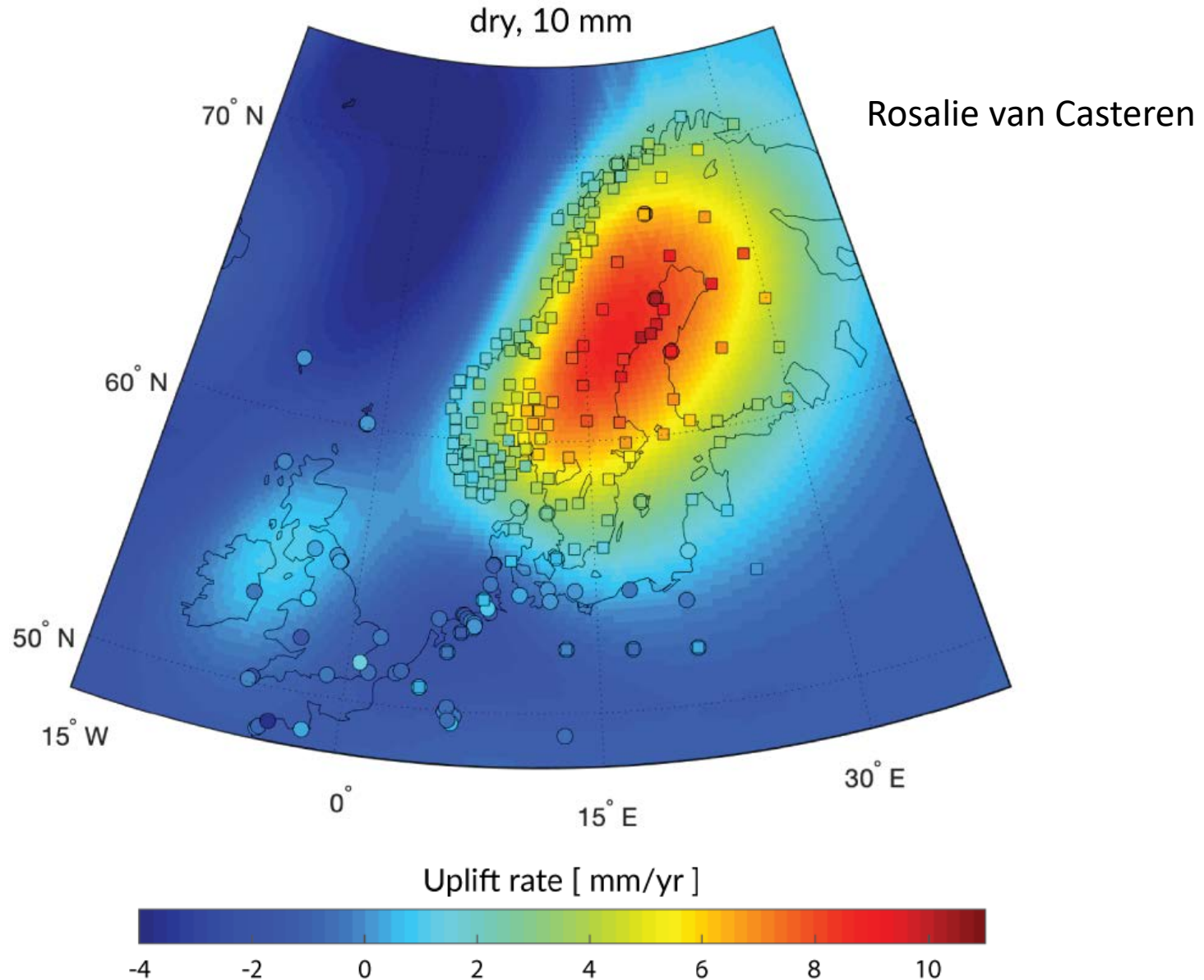


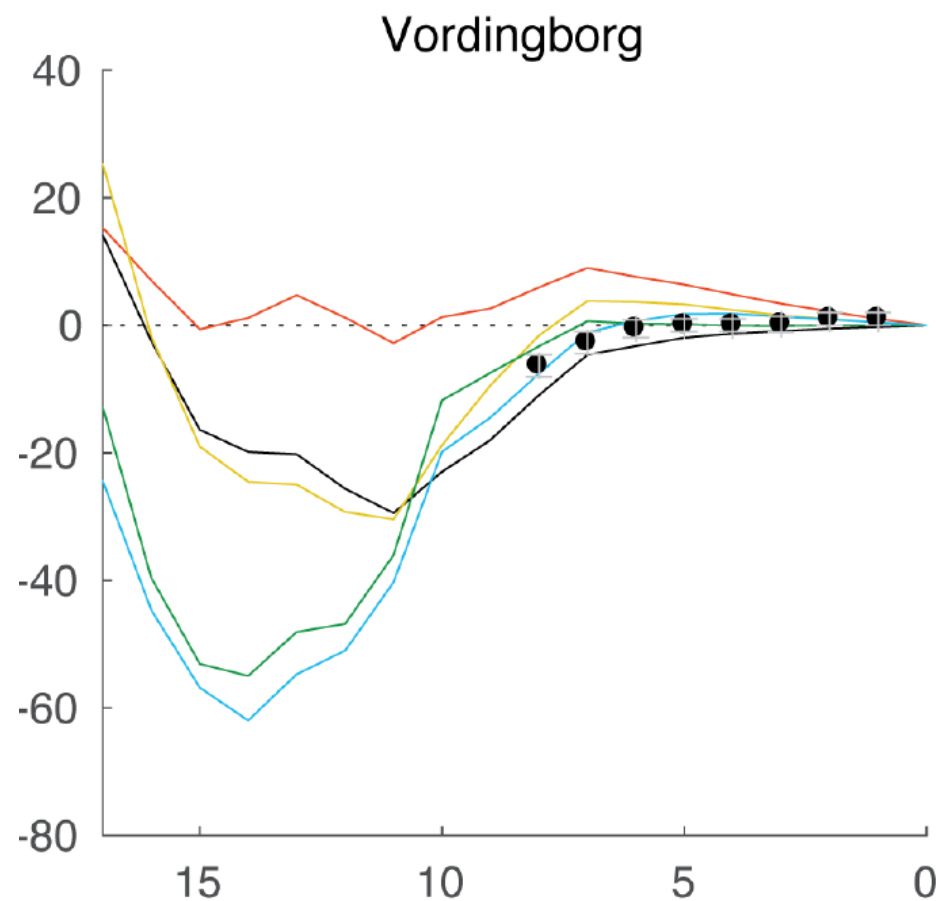
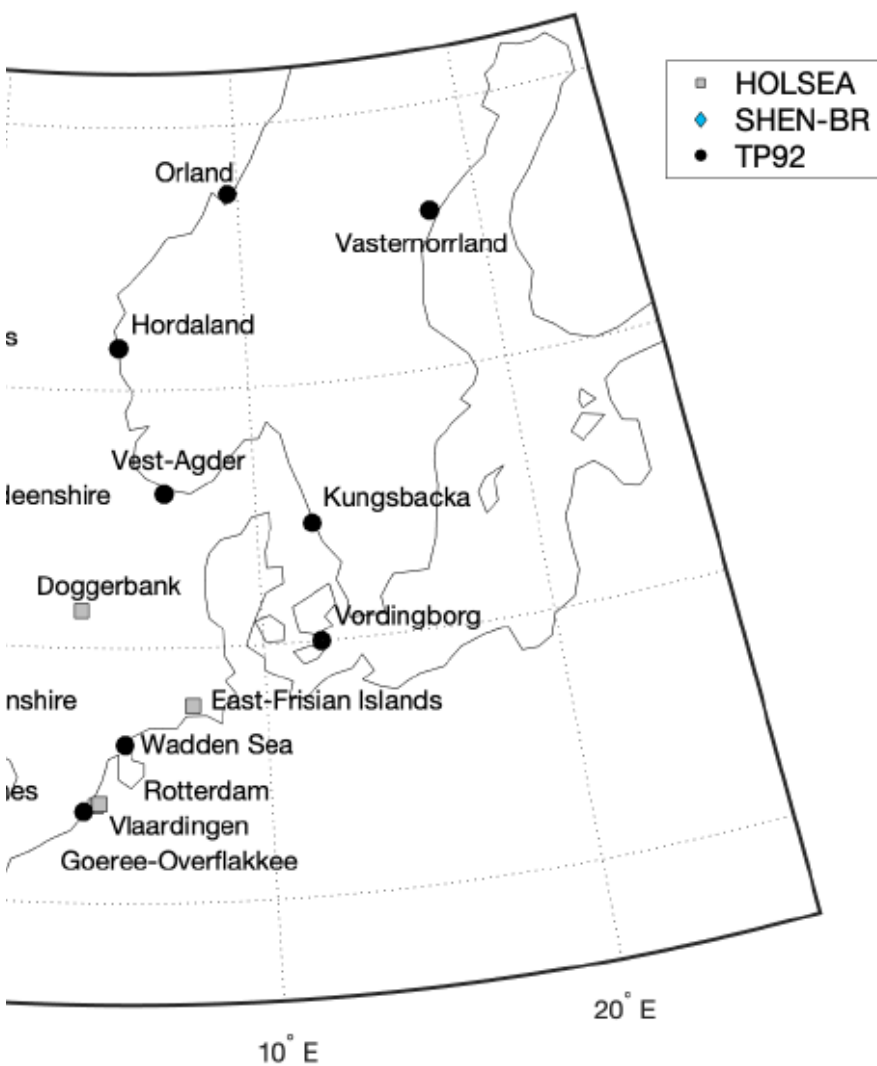
Whitehouse et al, in preparation

NW Europe - Temperature



NW Europe – Uplift rate



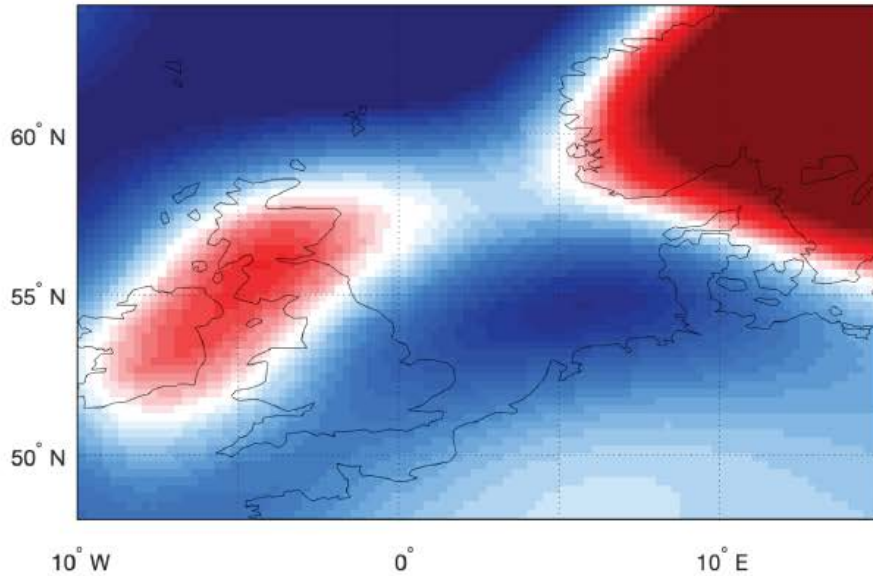


- VM5a BR
- 4mm dry BR
- 10mm dry BR
- 4mm wet BR
- 10mm wet BR

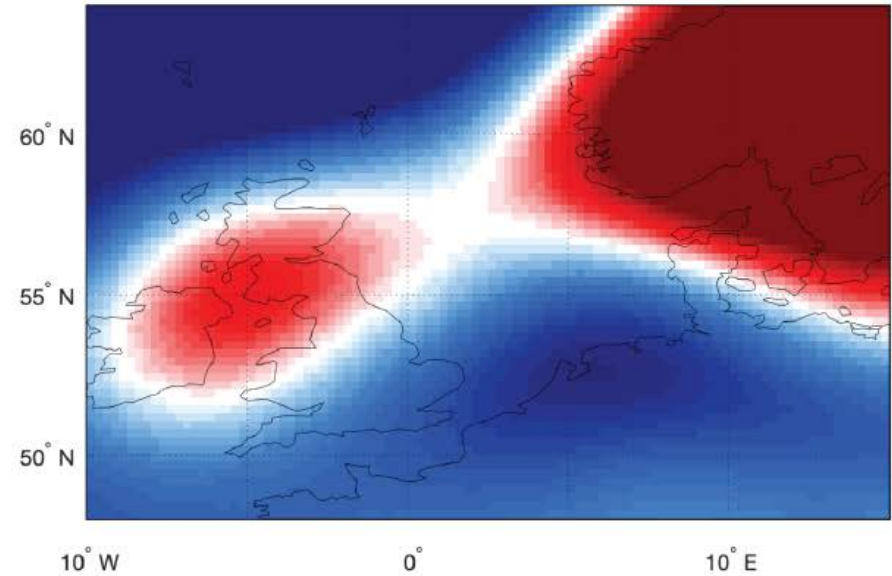
Rosalie van Casteren

NW Europe – RSL rate

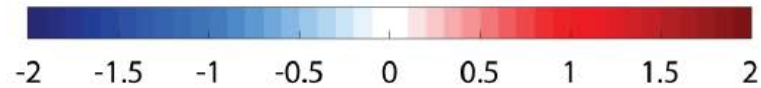
VM5a



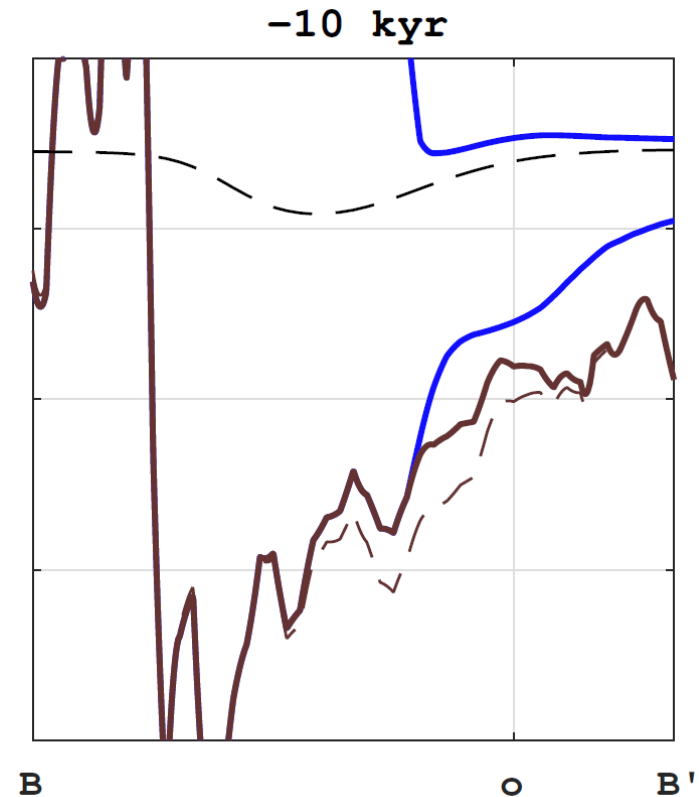
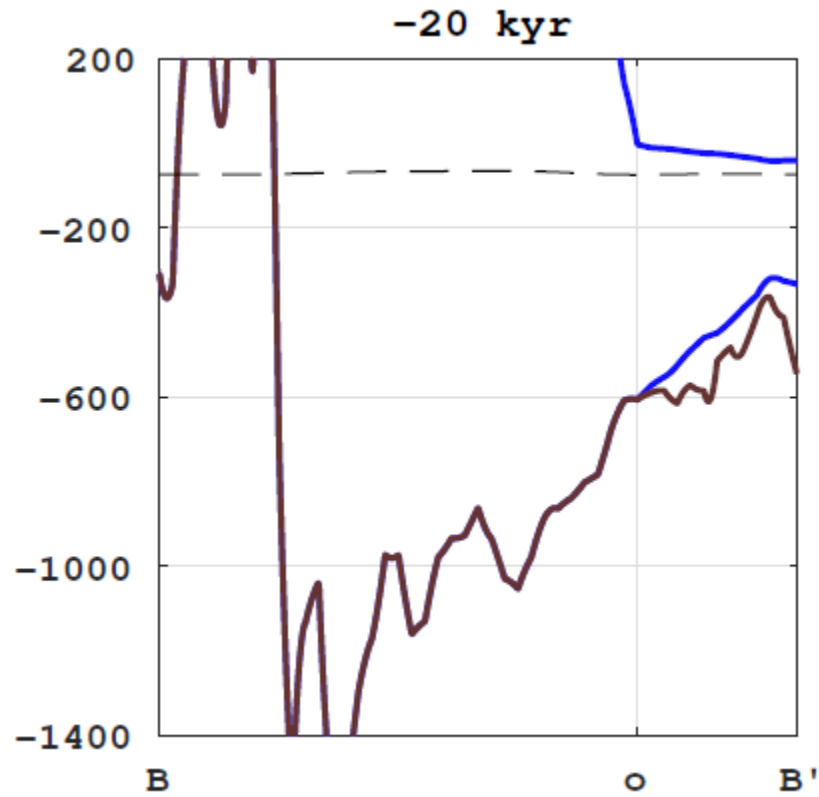
dry, 10 mm



Uplift rate [mm/yr]

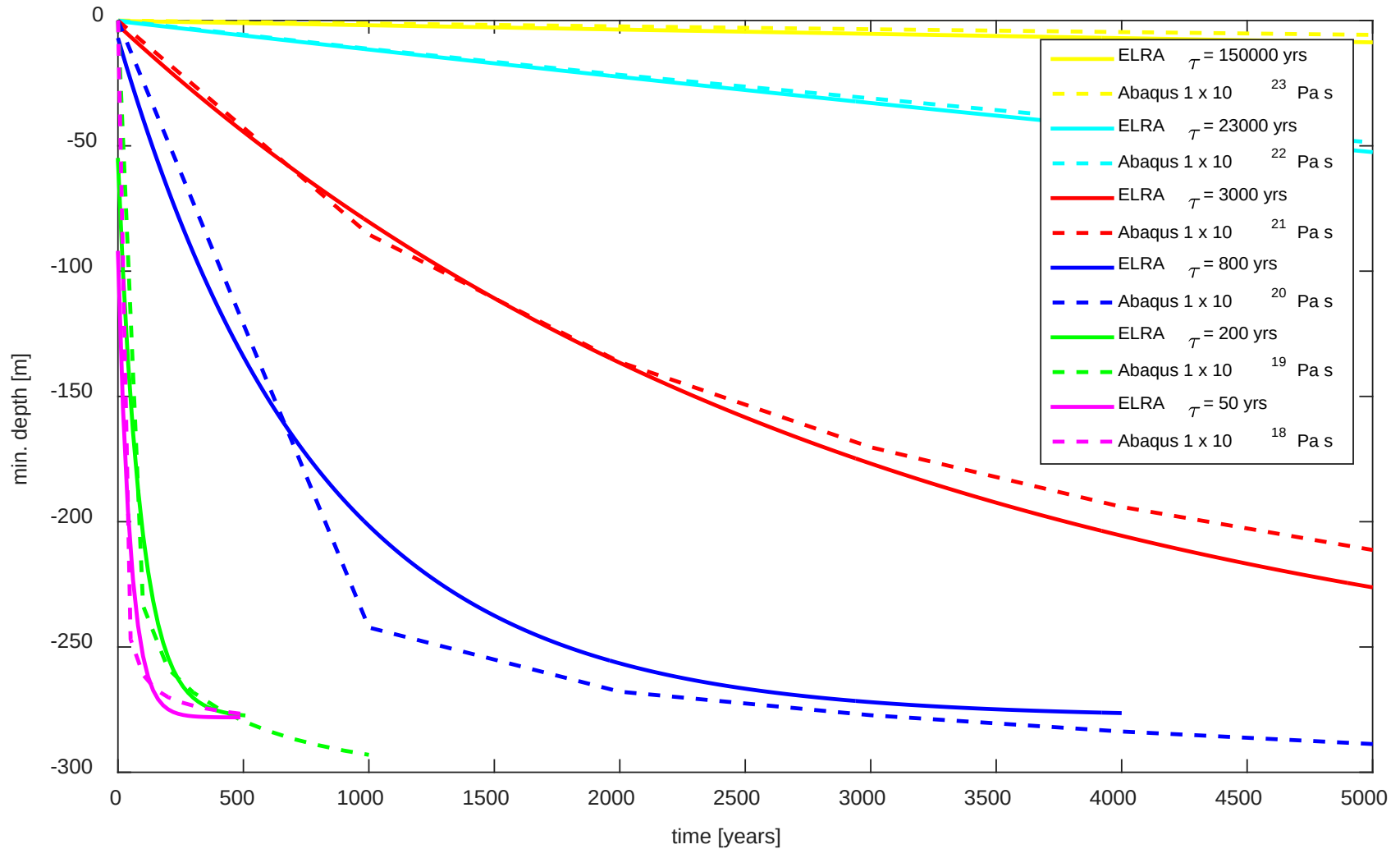


Solid Earth – Ice Dynamics Feedback

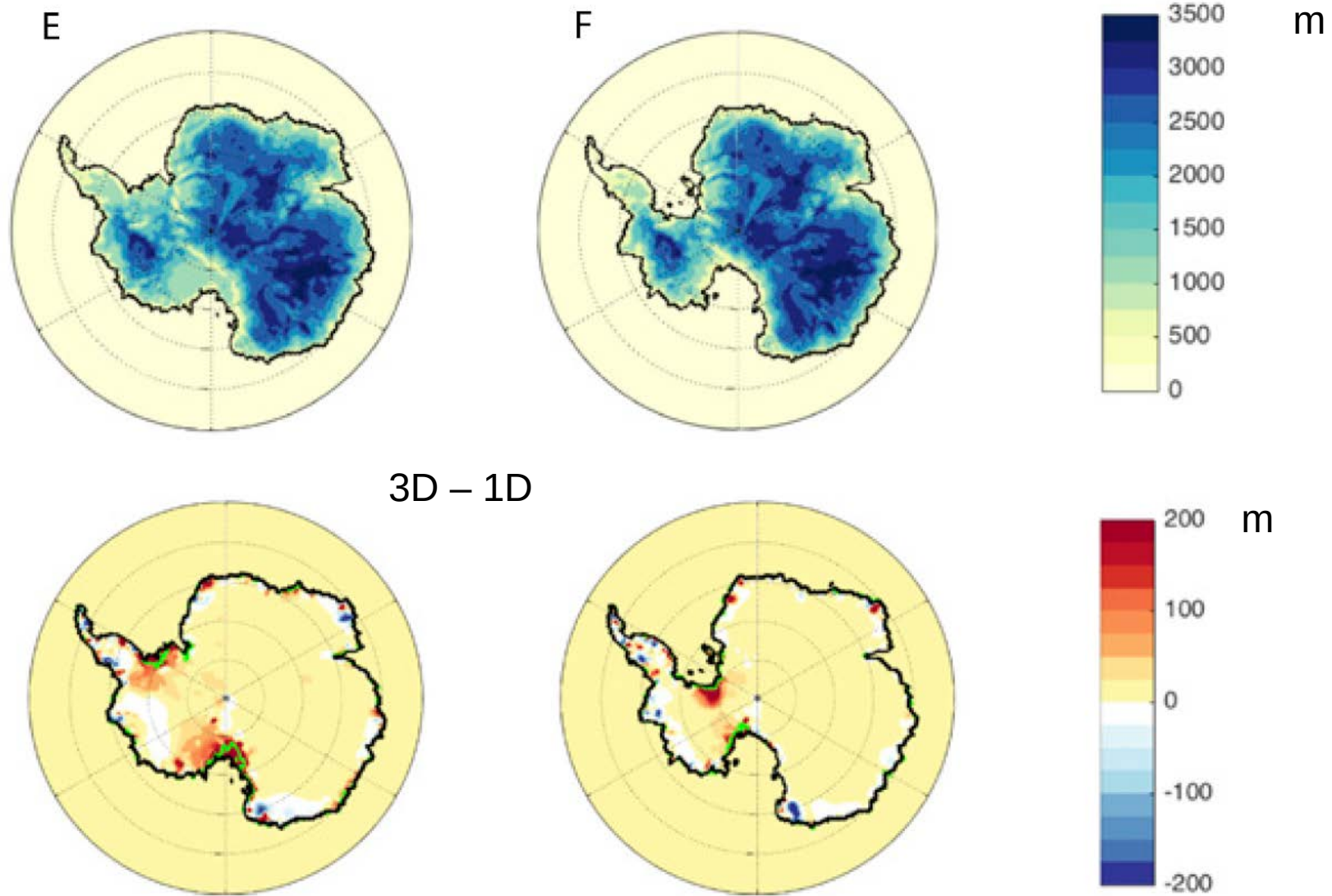


Dirk Oude Egbrink

Solid Earth – Ice Dynamics Feedback

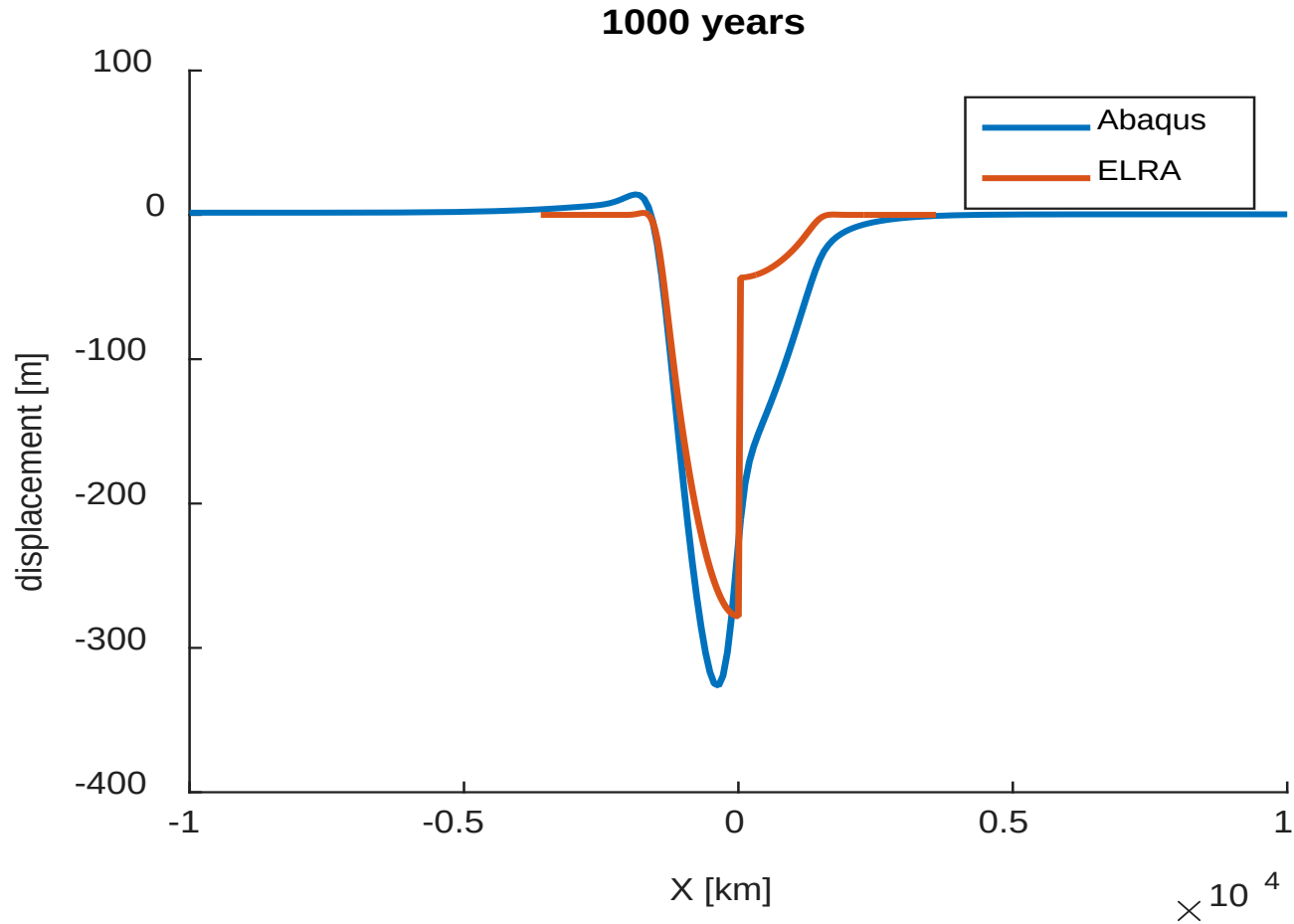


Solid Earth – Ice Dynamics Feedback



Gomez et al. (2018)

Solid Earth – Ice Dynamics Feedback



10^{19} Pa s

10^{21} Pa s

Open questions

Benchmark 3D models

Uncertainty estimates for 3D model predictions

Transient rheology

Coupling to ice dynamics

4D elastic parameters and density

References

Martinec (2000)

Wu (2004)

Latychev et al (2005)

A et al (2013)